



nextwork.org

Secure Packages with CodeArtifact



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The screenshot shows the AWS CodeArtifact console for the repository 'nextwork-devops-cicd'. A green banner at the top indicates that the repository was successfully created. Below the banner, the repository details are shown, including a description: 'This repository stores packages related to a Java web app created as a part of NextWork's CI/CD Pipeline series.' The 'Packages' section is expanded, showing a list of packages with columns for Package name, Namespace, Format, Latest version, Latest publish date, Publish status, and Upstream status. The packages listed are 'backport-util-concurrent', 'classworlds', 'google', and 'jstz05'.

Package name	Namespace	Format	Latest version	Latest publish date	Publish	Upstream
backport-util-concurrent	backport-util-concurrent	maven	3.1	4 minutes ago	Block	Allow
classworlds	classworlds	maven	1.1	4 minutes ago	Block	Allow
google	com.google	maven	1	4 minutes ago	Block	Allow
jstz05	com.google.code.findbugs	maven	2.0.1	4 minutes ago	Block	Allow



Introducing Today's Project!

In this project, we'll create a CodeArtifact repository to store your web app's packages!

Key tools and concepts

Services I used were Code artifact, IAM and EC2. Key concepts I learnt included least privilege policies and upstream repositories

Project reflection

This project took me approximately 2 hours. The most challenging part was when the authentication failed and had to create the IAM policy. It was most rewarding to get a build success at the end of the build.

This project is part three of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tomorrow.

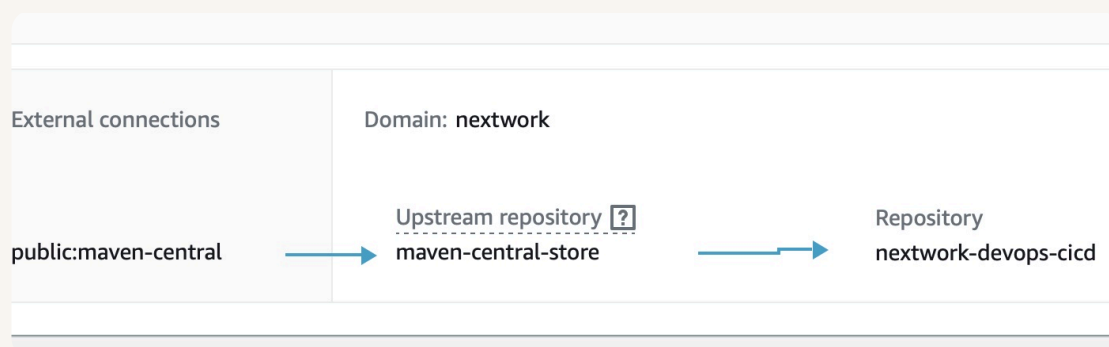


CodeArtifact Repository

CodeArtifact is a secure, central place to store all your software packages. Engineering teams use artifact repositories because it gives a consistent, reliable place to store and retrieve these components

A domain is like a folder that holds multiple repositories belonging to the same project or organization. We like using domains because they give you a single place to manage permissions and security settings that apply to all repositories inside it

A CodeArtifact repository can have an upstream repository. Upstream repositories are like backup libraries that your primary repository can access when it doesn't have what you need.





CodeArtifact Security

Issue

To access CodeArtifact, we need an IAM policy to give permission. I ran into an error when retrieving a token because by default, my EC2 instance doesn't have permission to access other AWS services (including CodeArtifact).

Resolution

To resolve the error with my security token, I created an IAM policy and Role and attached it to my instance. This resolved the error because the authorization token is like a temporary ID badge for your build tools to access CodeArtifact.

It's security best practice to use IAM roles because it avoids accidental leaks and enforces the principle of Least Privilege



The JSON policy attached to my role

The JSON policy I set up grants permission to get authentication tokens, find repository locations, and read packages from repositories.

Policy editor Visi

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": [  
7         "codeartifact:GetAuthorizationToken",  
8         "codeartifact:GetRepositoryEndpoint",  
9         "codeartifact:ReadFromRepository"  
10      ],  
11      "Resource": "*"   
12    },  
13    {  
14      "Effect": "Allow",  
15      "Action": "sts:GetServiceBearerToken",  
16      "Resource": "*",  
17      "Condition": {  
18        "StringEquals": {  
19          "sts:AWSServiceName": "codeartifact.amazonaws.com"  
20        }  
21      }  
22    }  
23  ]  
24 }
```



Maven and CodeArtifact

To test the connection between Maven and CodeArtifact, I compiled my web app using settings.xml

The settings.xml file is Maven's control center - it's where you tell Maven how to behave across all your projects. By configuring settings.xml properly, we're creating a seamless connection between Maven and CodeArtifact.

Compiling is like translating your project's code into a language that computers can understand and run. When you compile your project, you're making sure everything is correctly set up and ready to turn into a working app.

```
<settings>
<servers>
  <server>
    <id>nextwork-nextwork-devops-cicd</id>
    <username>aws</username>
    <password>${env.CODEARTIFACT_AUTH_TOKEN}</password>
  </server>
</servers>
<profiles>
  <profile>
    <id>nextwork-nextwork-devops-cicd</id>
    <activation>
      <activeByDefault>true</activeByDefault>
    </activation>
    <repositories>
      <repository>
        <id>nextwork-nextwork-devops-cicd</id>
        <url>https://nextwork-891377395566.d.codeartifact.us-east-1.amazonaws.com/maven/nextwork-dev
      </repository>
    </repositories>
  </profile>
</profiles>
```



Verify Connection

After compiling I checked my code artifact repository and I noticed that tons of new packages had been added .

us-east-1.console.aws.amazon.com

Successfully created nextwork-devops-cicd in nextwork.
Configure your package manager to install packages from nextwork-devops-cicd. [View connection instructions](#)

nextwork-devops-cicd [Info](#) [Delete repository](#) [Apply repository policy](#) [Edit](#)

Repository This repository stores packages related to a Java web app created as a part of NextWork's CI/CD Pipeline series.

Details
Domain, policy, tags, ARN, and upstream repositories.

Packages [Info](#) [Delete package](#) [View connection instructions](#)

Filter by package name prefix, format, namespace prefix, and origin controls

	Package name	Namespace	Format	Latest version	Latest publish date	Publish	Upstream
☐	backport-util-concurrent	backport-util-concurrent	maven	3.1	4 minutes ago	Block	Allow
☐	classworlds	classworlds	maven	1.1	4 minutes ago	Block	Allow
☐	google	com.google	maven	1	4 minutes ago	Block	Allow
☐	jst305	com.google.code.findby	maven	2.0.1	4 minutes ago	Block	Allow

Open "https://us-east-1.console.aws.amazon.com/codesuite/codeartifact/q/891377395566/nextwork/f/nextwork-devops-cicd/edit?region=us-east-1" in a new tab 5, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)



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